

Assignment 4

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Problem Statement

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Appendix

Part A

Add table and pie chart images.

What can you conclude about the effects of fitness scaling on selection pressure?

- Adding constants raises all fitness all values by that amount. So it smooths out all the values. For example, when we compare f_2 (which only squares X) and f_4 (which squares X and adds 20), we can see that the probabilities of f_4 are closer together. Probabilities are increased and reduced, so there is not such a gap between them, which also reduces selection pressure. - Multiplying by a constant does not change the selection probabilities. If we compare f_2 and f_3 , we can see that the probabilities stay the same even if we scale it. So scaling fitness does not have an effect on the selection pressure.